* variance

easy level

1) 5, 7, 9, 11, 13

	Mean	$(x-\bar{x})$	$(x-\bar{x})^2$
5	9	-4	16
7	9	-2	4
9	9	0	0
11	9	2	4
13	9	4	16
			40

1) $\frac{45}{5} = \boxed{9}$

1 Mean = 9
2 population = 8
4 sample var = 10
5 stand Devi = 3.16
3 population stan Devi = 2.83

2) Population

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$$

$$= \frac{40}{5} \quad \boxed{8}$$

3) Population stand

$$\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$$

$$= \sqrt{\frac{40}{5}}$$

$$= \sqrt{8}$$

$$= \boxed{2.83}$$

4) sample

$$s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$$

$$= \frac{40}{5-1} = \frac{40}{4} = \boxed{10}$$

5) Stand devi

$$\sqrt{10} = \boxed{3.16}$$

14

2)

x	$\bar{x}$	$x-\bar{x}$	$(x-\bar{x})^2$
10	30	-20	400
20	30	-10	100
30	30	0	0
40	30	10	100
50	30	20	400
			1000

Mean = 30
population = 200
p. stan devi = 14.14
Sample var = 250
Stan Devi = 15.81

① $\frac{150}{5} = \boxed{30}$

② $\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$

$\frac{1000}{5} = \boxed{200}$

③ $s^2 = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

$= \sqrt{\frac{1000}{5}}$

$= \sqrt{200}$

$= \boxed{14.14}$

④ $s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$

$= \frac{1000}{5-1}$

$= \frac{1000}{4}$

$\boxed{250}$

⑤

$s = \sqrt{250}$

$= \boxed{15.81}$

15

3)

x	$\bar{x}$	$x-\bar{x}$	$(x-\bar{x})^2$
2	6	-4	16
4	6	-2	4
6	6	0	0
8	6	2	4
10	6	4	16
			40

Mean = 6
population = 8
p. stan devi = 2.83
sample var = 10
stan Devi = 3.16

① $\frac{30}{5} = \boxed{6}$

② $\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$

$\frac{40}{5} = \boxed{8}$

③ $s^2 = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

$\sqrt{\frac{40}{5}}$

$\sqrt{8}$

$= \boxed{2.83}$

④ $s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$

$\frac{40}{5-1}$

$\frac{40}{4} = \boxed{10}$

⑤ $s = \sqrt{10}$

$= \boxed{3.16}$

Median & hard

4)

x	$\bar{x}$	$x-\bar{x}$	$(x-\bar{x})^2$
12	30	-18	
15	30	-15	
18	30	-12	
21	30	-9	
24	30	-6	
90	30	-60	

Mean = $\frac{180}{}$ 30
population =
p.stan devi =
sample var =
stan Devi =

1) $\frac{180}{6} = \boxed{30}$

4)

x	$\bar{x}$	$x-\bar{x}$	$(x-\bar{x})^2$
12	29.57	-17.57	308.70
15	29.57	-14.57	212.28
18	29.57	-11.57	133.86
21	29.57	-8.57	73.44
24	29.57	-5.57	31.02
27	29.57	-2.57	6.60
90	29.57	60.43	3651.785
			4417.665

Mean = 29.57
population = 631.09
pstan devi = 25.12
Sample var = 736.28
stan Devi = 27.13

① $\frac{207}{7} = \boxed{29.57}$

② $\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$

$\frac{4417.665}{7}$

$= \boxed{631.09}$

③ $s^2 = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

$\sqrt{\frac{4417.665}{7}}$

$\sqrt{631.09}$

$= \boxed{25.12}$

④ $s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$

$\frac{4417.665}{7-1}$

$\frac{4417.665}{6}$

$= \boxed{736.28}$

⑤ $s = \sqrt{736.28}$

$= \boxed{27.13}$